

PROGRAM OUTCOMES & PSOs MAPPING

Curricular Alignment & Engineering Competency Analysis — VibeGuard v3.0

TEAM NAME

TEAM SPARK

INSTITUTIONSreenidhi Institute of Science
and Technology**PROJECT NAME**

VibeGuard v3.0

FRAMEWORK

NBA / AICTE OBE Criteria

TEAM MEMBERS:

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Curricular Context: This document demonstrates how the design, algorithmic synthesis, and production implementation of **VibeGuard v3.0** rigorously satisfy the 12 Graduate Attributes (Program Outcomes, PO1–PO12) prescribed by the National Board of Accreditation (NBA) and ABET, along with discipline-specific Program Specific Outcomes (PSOs) in Computer Science, Cyber Security, and Software Engineering.

01 Program Outcomes (POs) Correlation Matrix

CODE	GRADUATE ATTRIBUTE / PO NAME	CORRELATION	KEY EVIDENCE IN VIBEGUARD ENGINEERING LIFECYCLE
PO1	Engineering Knowledge	HIGH (3)	Applied operating systems principles (IPC, Windows PATH, process control), graph/tree representations in Git commit refs, and algorithmic string searching in Rust.
PO2	Problem Analysis	HIGH (3)	Identified the Shift-Left Security Gap where centralized CI/CD detects secret leaks too late; modeled mathematical risk scoring: $\max(0, 100 - (15C + 8H + 3M + 1L))$.
PO3	Design / Development of Solutions	HIGH (3)	Architected a dual-engine architecture marrying Go's network orchestration with Rust's bare-metal file walk speeds; created fail-closed pre-push security firewall.
PO4	Conduct Investigations of Complex Problems	HIGH (3)	Benchmarked disk-staged tar archive vs stdout pipe deadlocks on Windows; profiled OSV API latency and engineered 10-worker pool achieving >80% speedup.
PO5	Modern Tool Usage	HIGH (3)	Leveraged Go 1.21+, Rust Cargo 2021, Git internals (hooks, rev-list, archive), MSVC linkers, PowerShell scripting, and Google Open Source Vulnerabilities (OSV) API.
PO6	The Engineer and Society	HIGH (3)	Protects digital infrastructure and user privacy by preventing catastrophic enterprise data breaches, credential leaks, and supply chain poisoning before code exposure.
PO7	Environment and Sustainability	MEDIUM (2)	Compiled native binaries (Go/Rust) consume near-zero CPU idle cycles (<100ms scan time), drastically reducing cloud CI/CD server compute hours and carbon emissions.
PO8	Ethics	HIGH (3)	Adheres to non-invasive privacy principles: 100% on-machine source code analysis; only anonymized package names and versions query OSV; zero code leaves local machine.
PO9	Individual and Team Work	HIGH (3)	Distributed cross-functional development across Rust engine engineering, Go hook orchestration, API integration, and validation test pipelines.
PO10	Communication	HIGH (3)	Designed clean, non-messy terminal reporting with grouped tables and color-coded severities; authored complete technical architecture and user

CODE	GRADUATE ATTRIBUTE / PO NAME	CORRELATION	KEY EVIDENCE IN VIBEGUARD ENGINEERING LIFECYCLE
			documentation.
PO11	Project Management and Finance	HIGH (3)	Automated deployment via setup.bat and build.bat; eliminated expensive enterprise SAST subscription costs through zero-cost open-source technology.
PO12	Life-Long Learning	HIGH (3)	Adapted to evolving language ecosystems (Rust 2021 edition memory safety, Go goroutine pooling) and real-time threat intelligence data schemas.

02 In-Depth Program Outcome Justifications

PO1: Engineering Knowledge & Mathematical Formulation

LEVEL 3

Apply mathematics, science, engineering fundamentals, and computing specialization to the solution of complex engineering problems.

How VibeGuard Addresses PO1: VibeGuard applies computer systems engineering principles: operating system process virtualization, memory-safe compiled systems programming in Rust, and Git Directed Acyclic Graph (DAG) commit traversal. The risk evaluation engine incorporates a deterministic mathematical model: $\text{RiskScore} = \max(0, 100 - (15 \times \text{Critical} + 8 \times \text{High} + 3 \times \text{Medium} + 1 \times \text{Low}))$, ensuring a quantitative threshold for push gate blocking.

PO2: Problem Analysis & Shift-Left Paradigm

LEVEL 3

Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions.

How VibeGuard Addresses PO2: The team analyzed why enterprise secret management repeatedly fails in production: traditional cloud CI scans occur after commits are pushed, by which point secrets are permanently recorded in distributed Git logs. VibeGuard formulates an autonomous pre-push barrier that halts vulnerability propagation at the exact boundary of origin.

PO3: Design & Development of Solutions

LEVEL 3

Design solutions for complex engineering problems and design system components or processes that meet specified needs.

How VibeGuard Addresses PO3: Engineered a dual-engine architecture: (1) Go CLI coordinator managing hook lifecycles, Git archive extraction, and OSV REST networking; and (2) Rust high-speed worker scanning ASTs and token regexes. Designed a fail-closed gate ensuring push aborts if either binary is missing or if risk exceeds threshold.

PO4: Investigations of Complex Problems

LEVEL 3

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data.

How VibeGuard Addresses PO4: During Windows testing, standard CLI pipe redirection caused deadlocks on large git archives. The team conducted empirical experiments, proving that disk-staged pure Go archive/tar processing eliminated OS pipe buffer blocks. Further benchmarked serial vs concurrent OSV lookups, reducing query time from 8.2s to 420ms.

PO5: Modern Tool Usage

LEVEL 3

Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools.

How VibeGuard Addresses PO5: VibeGuard integrates modern toolchains: Go 1.21+ toolchain, Rust Cargo compiler, MSVC toolset, Git internal hook subsystem, Google OSV API, and PowerShell system configuration scripts.

PO8: Professional Ethics & Data Privacy

LEVEL 3

Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

How VibeGuard Addresses PO8: Guarantees absolute client-side code confidentiality. Proprietary source code is never transmitted to external clouds; only anonymized package metadata queries the OSV API. Discovered secrets are masked in memory (first 8 characters followed by ***) before output generation.

03 Program Specific Outcomes (PSOs) Alignment

For Computer Science & Engineering, Cyber Security, and Information Technology disciplines, VibeGuard directly satisfies core PSOs:

PSO CATEGORY	PROGRAM SPECIFIC OUTCOME STATEMENT	VIBEGUARD CONCRETE IMPLEMENTATION & DEMONSTRATION
PSO 1 (Systems & Security)	Secure Software Engineering & Threat Mitigation: Ability to design, analyze, and build secure software architectures resilient against cyber threats, vulnerabilities, and malicious injections.	VibeGuard provides an active defensive barrier against OWASP Top 10 vulnerabilities (SQL injection, command injection, unsafe eval, insecure TLS) and supply-chain exploits (compromised PyPI/npm/Go dependencies) by intercepting flaws prior to deployment.
PSO 2 (High-Performance Computing)	Algorithmic Efficiency & Systems Programming: Ability to write optimized, high-throughput systems-level applications utilizing compiled languages and concurrency models.	Implemented the core scanner in Rust using memory-safe, zero-cost abstractions (walkdir, streaming regex automata), achieving sub-100 millisecond execution over 50+ files without runtime garbage collector stalls.

04 Outcome Attainment Summary

Assessment Conclusion: VibeGuard v3.0 achieves **Level 3 (Substantial / High Attainment)** across 9 of 12 Program Outcomes and both Program Specific Outcomes (PSO 1 & PSO 2). The project demonstrates holistic engineering readiness—spanning theoretical analysis, real-world systems software development, ethical data protection, and high-performance concurrent software engineering.